

Laurence Wilkes | Curriculum Vitæ

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Experience

KU Leuven

Leuven, BE

Postdoctoral researcher

Apr 2025–Sep 2025

- Developed algorithms for fast function approximation, utilising a novel set of sample points which produces an n -fold speed up over other algorithms.
- Produced high-performance FFT-based implementations of the algorithms introduced in the project, enabling large-scale parallelised numerical studies.
- Peer reviewer for multiple leading journals in numerical analysis.

KU Leuven

Leuven, BE

PhD researcher

Apr 2021–Mar 2025

- Research in function approximation and quasi-Monte Carlo methods, with emphasis on high-dimensional approximation.
- Presented at international conferences and seminars, including an invited talk at the Dagstuhl seminar on Algorithms and Complexity for Continuous Problems.
- Published results in international journals including the Journal of Complexity and peer-reviewed conference proceedings.
- Teaching Assistant for the Complexity Theory course; led exercise sessions and co-developed problem sets with solutions.

University of Warwick

Coventry, UK

URSS Research Project

Jul 2019–Jul 2020

- Awarded a competitive URSS grant conducting research in Extremal Combinatorics and Optimisation.
- The work resolved a stability conjecture the proof of which was published in the European Journal of Combinatorics.

Education

KU Leuven

Leuven, BE

Doctorate in Computer Science (Numerical Mathematics)

2021–2025

PhD Thesis: High-Dimensional Approximation Using Generated Point Sets and Rank-1 Lattices

University of Warwick

Coventry, UK

*Integrated Bachelor's and Master's in Mathematics, **First class honours***

2015–2020

Dissertation: Stability of maximum sum-free sets in a lattice

University of Göttingen

Göttingen, DE

Erasmus+ year

2017–2018

Beauchamp College

Leicester, UK

A levels, Subjects: Maths, Further Maths, Electronics, Physics, Music

2013–2015

Technical skills

Programming:

Proficient in: C++, Julia, Python (NumPy, SciPy), L^AT_EX;

Experience with: C, Fortran, Java, MATLAB, JavaScript

Numerical mathematics:

Approximation theory, Numerical linear algebra, Optimisation methods, Least squares, Random matrices, PDEs and numerical solvers, Financial Engineering, DFTs and fast matrix operations, Iterative methods (Krylov, Conjugate gradients), Spectral methods

Systems, tools & workflows:

Git, Linux, SSH, CI/CD, unit testing, Vim, parallel processing (GPU, multi-threading)

Skills, interests and achievements

Languages: English (Native), German (B1/B2), Polish (A1/A2), Dutch (A2)

Music: French Horn (ABRSM Grade 8), Piano (ABRSM Grade 8), Folk Instruments

Sports: Skiing, Running, Trampolining, Cycling